

OS Scheduling & Context Switching

1. What I Understood

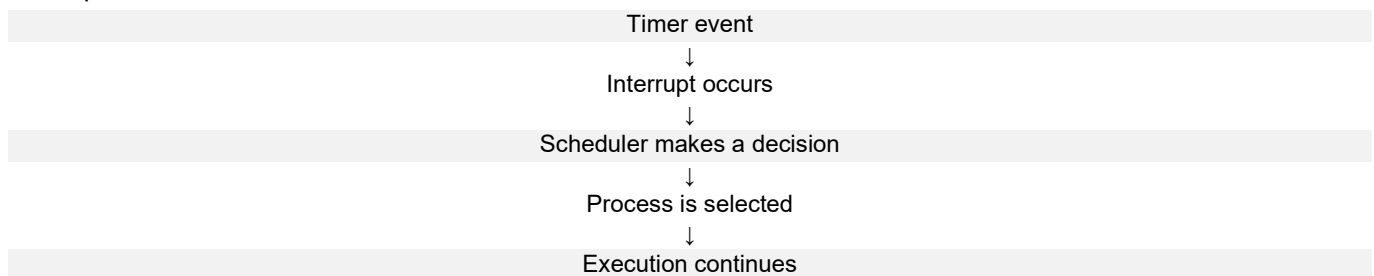
My main responsibility is OS Scheduling and Context Switching. I understood that when several processes are ready, the CPU needs a rule to decide which process should run. This decision is made by the scheduler.

Context switching means changing the CPU from one process to another. Before changing, the important information of the current process is saved so it can continue later.

2. Timer and Interrupts

A timer helps keep track of time. In our simulator, it can create an event when a fixed amount of time has passed. The timer does not choose the next process; it only helps start a scheduling decision.

An interrupt is a signal that tells the CPU that something needs attention. A timer overflow can generate a timer interrupt.



Simple Interrupt Flow

Main program → Interrupt → Identify source → Save required state → Handle interrupt → Resume execution.

3. CPU Scheduling

CPU scheduling means deciding which ready process should get the CPU next. The three scheduling methods connected to my work are FCFS, Round Robin and Priority Scheduling.

FCFS – First Come First Served

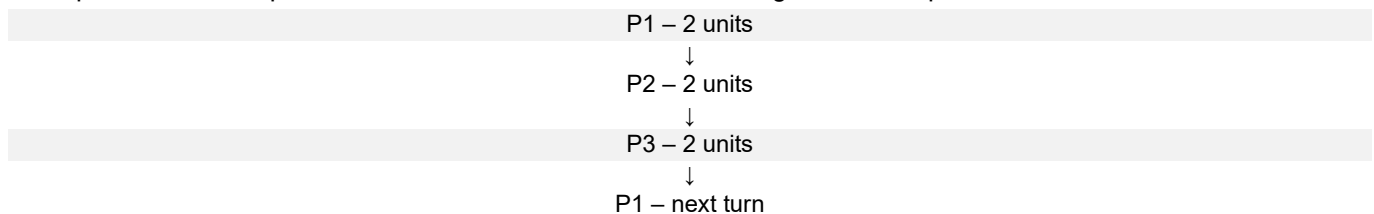
The process that arrives first gets the CPU first. Example: P1, P2 and P3 arrive in that order, so the execution order is P1 → P2 → P3.

A problem is that a long process at the front can make shorter processes wait. This is called the convoy effect.

Round Robin

Round Robin gives each process a fixed amount of CPU time called a time quantum. If the process is not finished, it goes back to the ready queue.

Example: with a time quantum of 2 units: P1 → P2 → P3 → P1 again, until all processes finish.



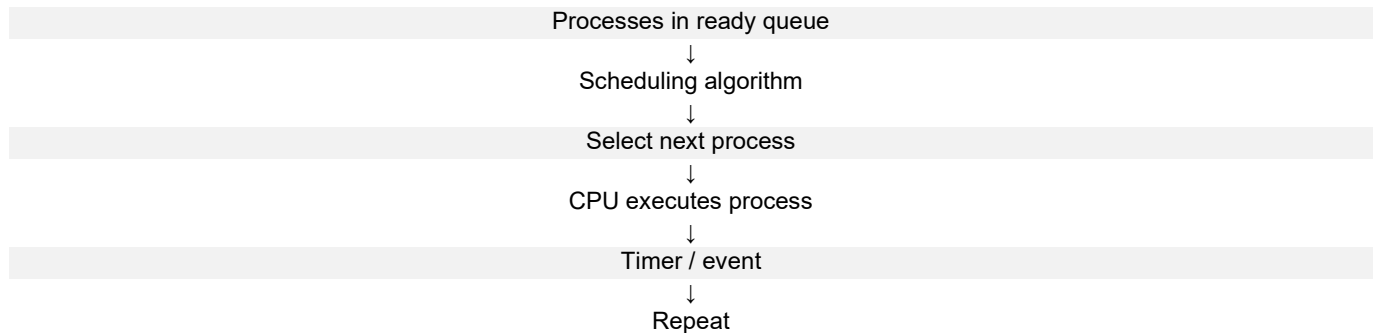
The timer is useful here because it can indicate when the time quantum has ended.

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Priority Scheduling

In Priority Scheduling, each process has a priority and the scheduler selects according to the project's priority rule. The exact priority convention should be fixed by the project implementation.

4. Scheduling Flow



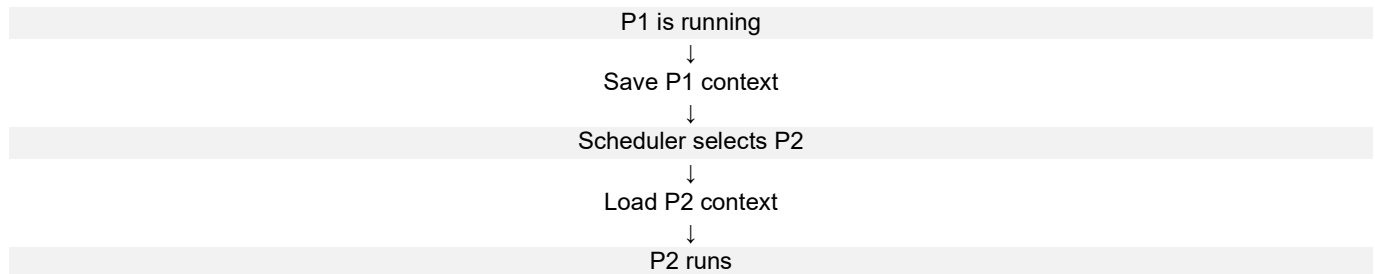
5. Context Switching

Context switching means moving the CPU from the current process to another process. The current process cannot simply be forgotten because it must continue from the correct point later.

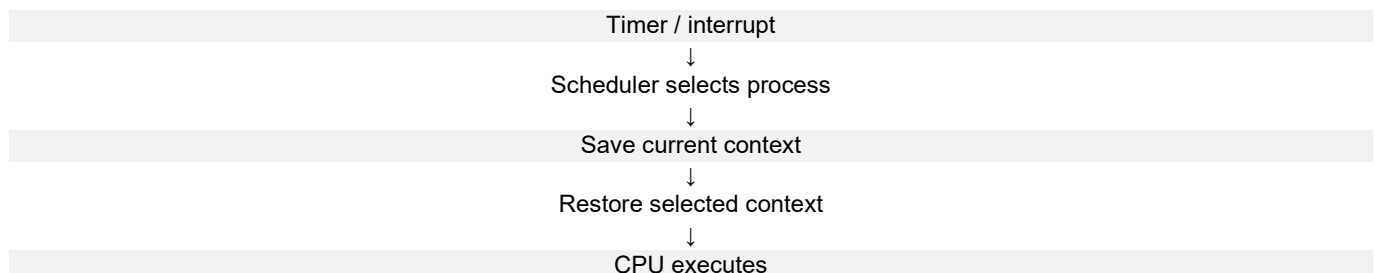
The important information is saved first, and then the next process information is restored. In our simple simulator this can include the Program Counter (PC), registers and status information.

Ex:

Suppose P1 is running and P2 is waiting. A timer event occurs and the scheduler selects P2. P1's information is saved, P2's information is loaded, and P2 starts or continues execution.



6. How Scheduling and Context Switching Work Together



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Simple Round Robin Ex:

P1 runs for 2 units → timer expires → scheduler selects P2 → context switch → P2 runs for 2 units → timer expires → next process is selected.

This shows the connection between the timer, interrupt, scheduler and context switch.

Decisions:

1. **Timer as a scheduling trigger.**
2. **Use the three assigned algorithms.**
3. **Save and restore process context.**